

BME 3301 “Introduction to Biomechanics Laboratory” Syllabus (Fall 2026)

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Class hours: Wednesday 3:30-5:20 pm

Office hours: 11 am M & W at J337
or by appointment

Location: J134

Course Content

This course provides some background theory to support real world laboratory applications to practice analyzing biomechanical concepts of human movement. Topics include; range of motion, introduction to motion capture methods, ground reaction forces, inverse dynamics, gait analysis of walking and running, and introduction to biomechanical modeling.

Course instructional format

The class will have short lectures, class activities, and group laboratories involving student participation. Students will be assessed based on participation, individual worksheet assignments, group laboratory reports and a final oral presentation.

Reference Book

Research Methods in Biomechanics 2nd Edition, Robertson et al. Human Kinetics, 2004. ISBN: 073603966X

Grading

- **10% Attendance and Class Participation**
 - It is important that students attend the lab on time. Missing days will hurt your grade in this course. Please contact the instructor for any and all anticipated absences, and especially emergencies relating to class attendance. The student will get an “F” after the second unexcused absence.
- **30% Individual Worksheets**
 - Students may discuss individual assignments with other students, but should write up solutions independently. Late submissions will not be accepted, unless previously arranged with the instructor.
- **40% Laboratory Reports**
 - Students may discuss laboratory reports with other students, but should write up solutions only with their groups. Late submissions will not be accepted, unless previously arranged with the instructor.
- **20% Final Research Presentation**
 - Self-selected student groups will present a chosen topic and answer questions from the instructor and other students. Additional details will be provided during the semester.

Advice

The assignments and laboratory preparation can consume a very large amount of time depending on your background and how well you have synthesized the course material and supplementary readings. You should prepare for the laboratory sessions prior to arriving to collect the data. This will insure that the data you collect will address the lab questions and limit the number of times you have to recollect the data. Make sure to start the assignments and lab preparation early, as there will be a new assignment frequently.

Grading Scale:

A	≥ 90%
A-	87-89.9%
B+	83-86.9%
B	80-82.9%
B-	77-79.9%
C+	73-76.9%
C	70-72.9%
C-	67-69.9%
D+	63-66.9%
D	60-62.9%
F	<60%

Attendance and Participation

Rubric

A = Fully engaged in class and team discussions. Asked and answered questions from instructor and classmates.

B = Mostly engaged in class and team discussions. Occasionally asked and answered questions from instructor and classmates. Occasionally distracted or engaged in other activities.

C = Only partially engaged in class discussions. Only responded when directly asked a question by the instructor. Occasionally distracted or engaged in other activities.

D = Rarely participated in class discussions. Only partially able to answer questions directly posed by professor. Frequently distracted or engaged in other activities.

F = Attended class but did not actively participate. Unable to answer questions directly posed by professor.

Participation Grade for Semester	Perfect Attendance	1 Absence	2+ Absences
A	10%	9%	Fail
B	8%	7%	Fail
C	7%	6%	Fail
D	6%	5%	Fail
F	5%	4%	Fail

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code, included here by reference (see http://www.ltu.edu/currentstudents/honor_code.asp). In general, any solution obtained from any source should be properly referenced. Specific policies for assignments in this course are:

- All undergraduate students are required to write the following pledge on all academic work submitted along with a signature. If the pledge does not accompany an assignment, I can choose to not grade it.
"I have neither given nor received unauthorized aid in completing this work, nor have I presented someone else's work as my own."
- All quizzes and exams must be by individual effort. Unprogrammed calculators are allowed, but use of laptops or programmed calculators as well as observation of others' work will be considered "cheating" as defined in the Honors Code. Plagiarism will not be tolerated.

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Course Objectives:

- Identify research tools in biomechanics and examples of the relevant fields that this data could be used in.
- Compute anthropomorphic averages and standard deviations for human body segments in different populations of individuals.
- Measure joint range of motion in three human anatomical reference planes.
- Compute center of gravity for human body segments and for the whole body in various static positions.
- Demonstrate the use of motion capture tools for the measurement of dynamic human movement activities.
- Measure the ground reaction forces, moments and center of pressure of a person using force and/or pressure sensors.
- Use biomechanical models to predict muscle contraction forces and simulate human movements.
- Produce effective written and verbal communication in the form of a team report on the designs that they developed.

Tentative Calendar

Lab	Date	Topics	Lab Assignment*	Equipment/Software
1a	8/26	Intro to Experimental Biomechanics	Segments Length	Measuring tape/Excel
1b	9/2	Planar Kinematics	Joints Range of Motion	Goniometers/Kinovea
2a	9/9	Functional Movement Screening	Squatting	Video/Kinovea
2b	9/16	Center of Gravity	Jumping	Video/Kinovea
3a	9/23	Motion Capture Methods	Neck Range of Motion	Vicon/Nexus
3b	9/30	Ground Reaction Forces	Balance	Force Plates/Nexus
4a	10/7	Gait Patterns	Walking	Vicon/Nexus
4b	10/14	Plug-in Gait Modelling	Running	Vicon/Nexus
5a	10/21	Markerless MoCap	Posture	OpenCAP
5b	10/28	Markerless MoCap	Ergonomics Design	OpenCAP
	11/4	Research Experimental Design	Human Biomechanics	Mocap
	11/11	Research Cont.	Collect Research Data	Mocap
	11/18	Research Cont.	Analyze Research Data	Mocap
6	12/2	Muscle	Optimization	OpenSIM
Final	TBA	Final Group Presentations		

* Groups will be assigned and details will be provided.